Complete Dosing Analysis — Revised & Confirmed

150 Gallon Reef | Based on ICP-MS MSR235401 (29.04.2026) | BRS 2-Part | pH Probe Recalibrated

Report: **May 02, 2026** | Reef Moonshiners Handbook Rev.5

Key findings from full analysis: (1) pH probe had drifted +0.35 units over ~1 year without calibration — true pH range is 8.09–8.42, which is perfectly normal. No pH crisis exists. (2) Ca/Alk imbalance was the real issue — Ca high (459) from overdosing, Alk low (6.8) from underdosing relative to coral consumption. (3) $NaHCO_3$ product is correct for this tank — no product change needed. (4) Refugium photoperiod is fine — no change needed. (5) All trace element findings from ICP-MS remain unchanged.

Section 1 — pH (Resolved)

Parameter	Meter Reported	True Value	Target	Status
Morning low	8.44	**8.09**	7.95–8.15	**✓ Normal**
Afternoon high	8.77	**8.42**	8.25–8.45	**✓ Normal**
Daily average	8.59	**8.24**	8.15–8.30	**✓ Textbook perfect**
Daily swing	0.33	**0.33**	0.2–0.35	**✓ Normal**

Probe drift confirmed at +0.35 units after ~1 year without calibration. Probe health is excellent — electrode slope 99.0%, consistent offset across range. Simple zero-point drift, not membrane degradation. Probe recalibrated and reading correctly. $NaHCO_3$ Alk product is appropriate for this pH range. Refugium photoperiod requires no change.

pH Probe Maintenance: Recalibrate every 4 weeks using fresh, sealed, in-date pH 7.00 and 10.00 buffers. Never reuse opened buffer solution. Store probe tip submerged in KCl solution between calibrations — never in RODI or dry. Replace probe if slope drops below 90% at next calibration.

Section 2 — Calcium & Alkalinity

Parameter	ICP Result	After adjustment	Target	Status
Alkalinity	6.8 dKH	**7.5 dKH**	7.8–8.5 dKH	**0.3 dKH to go**
Calcium	459 mg/L (high)	**408 mg/L**	420–440 mg/L	**22 mg/L below floor**

Calibrated Consumption Rates

Parameter	How calculated	Rate	Confidence
Ca consumption	51 mg/L drop over ~12 days with pump stopped	**4.25 mg/L per day**	Good — real measured data, ±10-15% for day uncertainty
Alk consumption	139 ml/day $NaHCO_3$ maintaining ~6.8 dKH (near-stable)	**0.80 dKH per day**	Good — confirmed by stable Alk under known dose

Section 3 — 2-Part Pump Settings

Phase 1 — Correction (Now → ~11 days)

Pump	Product	Rate	Net effect	Expected result
Alk pump	BRS $NaHCO_3$ (370g/gal stock)	**160 ml/day**	+0.12 dKH/day net gain	Reaches 7.8 dKH in ~3 days

Ca pump	BRS CaCl2 (200g/gal stock)	**185 ml/day**	+1.97 mg/L Ca/day net gain	Reaches 430 mg/L in ~11 days

Phase 2 — Maintenance (after both reach target)

Pump	Product	Rate	Delivers	Matches consumption
Alk pump	BRS NaHCO3	**139 ml/day**	0.798 dKH/day	0.80 dKH/day ✓
Ca pump	BRS CaCl2	**126 ml/day**	4.25 mg/L Ca/day	4.25 mg/L/day ✓

When to switch from Phase 1 to Phase 2: test Ca and Alk in 5 days. Expected at 5 days: Ca ~418 mg/L, Alk ~8.1 dKH. Switch to maintenance rates once Ca reaches 425–430 mg/L AND Alk is 7.8–8.5 dKH. Do not switch on both at the same time if one has not yet reached target.

Pump	Switch to maintenance when...	Maintenance rate
Alk pump	Alk reaches 7.8–8.0 dKH (expected ~3 days)	139 ml/day
Ca pump	Ca reaches 425–430 mg/L (expected ~11 days)	126 ml/day

Section 4 — Dosing Pump Recommendations (Trace Elements)

Priority	Element	Product	Daily Volume	Why pump
Essential	Alkalinity	BRS NaHCO3	139–160 ml/day	Continuous dosing prevents Alk swings — already on pump ✓
Essential	Calcium	BRS CaCl2	126–185 ml/day	Continuous dosing prevents Ca swings — already on pump ✓
High	Iron	RM Iron-X (10x)	1.7 ml/day	Macroalgae consumes Iron all day — continuous > single dose
High	Manganese NANO	RM Manganese NANO	40 ml/day	Large volume + macroalgae consuming continuously
High	Copper	RM Copper NANO	24 ml/day	Sensitive element — pump prevents spikes/troughs
High	Cobalt NANO	RM Cobalt NANO	10 ml/day	Crashed between tests — continuous supply prevents dips
Medium	Chromium NANO	RM Chromium NANO	22 ml/day	Small Classic volume hard to measure — NANO on pump easier
Medium	Selenium NANO	RM Selenium NANO	8 ml/day	Small Classic volume hard to measure — NANO on pump easier

Keep manual (no pump needed): Nickel 5.5 ml/day (syringe — no NANO product exists) | Vanadium 9 drops/day (cobalt-blue dropper bottle) | Iodine 3–5 drops/day (Seachem Reef Iodide ONLY, cobalt-blue dropper bottle — handbook specifies manual drop dosing for Iodine, not pump)

Section 5 — Complete Daily Dosing Plan

Dose dailies directly into display tank. Use individual syringes — never pre-mix elements. Iodine and Vanadium use cobalt-blue glass dropper bottles (1 drop = 0.03 ml). Dose corrections into sump/overflow.

Element	ICP 2 Result	Target	Current Dose	Notes
Iron (Classic or Iron-X)	0.05 µg/L	0.70–1.0	**17–18 ml/day (Classic) or 1.7 ml/day (Iron-X)**	Still depleted. Macroalgae consuming ~0.95 µg/L/day. Increase gradually. Iron-X (10x concentrate) recommended for pump use.
Copper (NANO)	0.07 µg/L	0.40–0.60	**24 ml/day**	Crashed between tests. Consumption surged. Once at 0.50, reduce to ~19.5 ml/day maintenance.
Cobalt (NANO)	0.11 µg/L	0.40–0.60	**10 ml/day**	Crashed between tests. Need ~10 ml/day NANO to recover.
Iodine (Seachem Reef Iodide)	44.5 µg/L	75–95	**3–5 drops/day**	Dropped from 92.5 to 44.5. Start immediately. Seachem Reef Iodide ONLY. Cobalt-blue dropper bottle. Adjust from ICP to ICP. If below 30 on next test, double drops.
Vanadium (drops)	0.54 µg/L	1.0–2.0	**9–10 drops/day**	5 drops only maintaining 0.54. Need ~9-10 drops to approach 1.0 target.
Manganese (NANO)	0.11 µg/L	0.40–0.60	**40 ml/day**	First detection. Continue increasing. ICP-MS may not detect reliably even when dosed — next ICP will guide.
Chromium (Classic)	0.37 µg/L	0.35–0.55	**1.7 ml/day**	In target but slowly dropping. Slight increase from 1.6 ml/day.
Selenium (Classic)	0.278 µg/L	0.35–0.55	**0.6 ml/day**	Just below target. Once at 0.40–0.45, reduce to 0.30–0.35 ml/day.
Nickel (Classic)	1.56 µg/L	2.5	**2.67 ml/day ×2 days then 5.5 ml/day**	Correction: 2.67 ml/day for 2 days (spreadsheet calculated). Then maintenance at 5.5 ml/day. Previous 5.0 ml/day maintained 2.12 µg/L — consumption increased.

Section 6 — Pending ICP Correction Doses

Dose into sump/overflow. Allow 1 minute between elements. Never pre-mix.

Element	Current	Target	ml/Day	Days	Product
Barium	5.68 µg/L	30 µg/L	34.52	4	RM Barium
Strontium	5.3 mg/L	10 mg/L	9.88	5	RM Strontium UHP
Fluoride	0.66 mg/L	1.5 mg/L	53.00	9	RM Fluoride
Potassium	398 mg/L	410 mg/L	139.06	1	Brightwell Potassion-P (100g/L RODI)
Magnesium	1322 mg/L	1350 mg/L	381.98	1	BRS Mag-Mix ONLY (400g/L RODI)
Nickel	1.56 µg/L	2.5 µg/L	2.67	2	RM Nickel
Caesium	1.34 µg/L	5 µg/L	1.04	2	RM Caesium
Zinc	0.19 µg/L	5 µg/L	0.91	3	RM Zinc (monthly routine)
Molybdenum	13.8 µg/L	15 µg/L	6.81	1	RM Molybdenum

Section 7 — Monitoring & Next Steps

In 5 days	Test Ca and Alk with home test kits. Expected: Ca ~418 mg/L, Alk ~8.1 dKH. If Alk already at 7.8+: reduce NaHCO3 back to 139 ml/day. Confirm Ca is trending upward.
In ~11 days	Ca should be near 430 mg/L. Reduce CaCl2 pump to 126 ml/day maintenance. Test both Ca and Alk to confirm both in target range simultaneously.
In 2 weeks	Send next ICP-MS sample. This will be the key calibration test — with Ca/Alk balanced, expect Strontium and Barium corrections to hold better. Also confirms Iron, Copper, Cobalt response to increased doses. Take sample before daily dosing as usual.
Monthly	Recalibrate pH probe using fresh sealed buffers (pH 7.00 and 10.00). Store probe tip in KCl solution. Check Alk and Ca with test kit and adjust pump rates if drifting.
Each ICP cycle	Recalibrate all daily doses from actual ICP results — especially Iron, Copper, Cobalt and Vanadium which showed highest consumption variance. Zinc should be treated as a monthly correction element given high consumption rate. Once Rubidium falls below 300 µg/L begin monthly Rubidium maintenance dosing.

Report: May 02, 2026 | Based on ICP-MS MSR235401 (29.04.2026) | 150 gal | Large coral load + macroalgae | BRS 2-part system | pH probe recalibrated — true range 8.09–8.42 (normal). Ca/Alk consumption rates calibrated from measured data. Per Reef Moonshiners Handbook Rev.5. All dosing recommendations should be verified against next ICP results.